

Doom PWAD QA Report

Run 105b84e5-b576-40c1-8e68-4b4e9e472044 | Map E1M1 | Outcome stuck

Executive Summary

This is an autonomous QA test of E1M1 (DRKWRLD1) using an LLM agent to validate map integrity and playability. The test concluded with a 'stuck' outcome after 58 seconds due to critical navigation failures preventing basic gameplay progression.

Pass/fail: map_navigation: FAIL combat_engagement: FAIL resource_balance: FAIL secret_coverage: FAIL
overall_verdict: FAIL

- The agent was unable to move beyond the starting position, resulting in a navigation softlock.
- No combat was engaged due to the player's inability to move or acquire a weapon.
- Resource balance could not be assessed as the agent could not reach any map items.
- No secrets were discovered because the agent was trapped at the spawn location.

No technical problems encountered during this test run; all 10 agent actions were successfully processed by the LLM.

Test Item And Environment

The test evaluated map E1M1 using the Freedoom1 IWAD on skill level 1. The map contains 69 total enemies (25 at this skill level), 78 items, 3 keys, and 8 secret sectors within a 5952x3712 unit footprint. The static complexity is high, featuring a significant percentage of hitscanner enemies.

The test was performed using ViZDoom as the game engine, with the gemini-3.1-flash-lite LLM model orchestrating actions via the MCP tool interface. The run was executed at difficulty level 1 with a max_ticks cap of 500.

Metric	Value	Metric	Value
Status	completed	Duration	58
Actions	10	LLM calls	10
Final HP	100	Kills	0
Position samples	22	Movement units	45.8
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/105b84e5-b576-40c1-8e68-4b4e9e472044.mp4		

Static Analysis Context

Metric	Raw WAD	Difficulty 1	Notes
Enemies	69	25	44 hidden by skill/multiplayer flags
Items	78	71	7 hidden by skill/multiplayer flags
Secrets	8	8	Sector special 9
Geometry	571 linedefs	99 sectors	5952 x 3712 units

Selected-difficulty estimate: hard. Health ratio: 0.1909. Ammo ratio: 0.1848.

Coverage Evaluation

Coverage was negligible, limited entirely to the starting position. The agent explored zero sectors, engaged zero enemies, and found zero secrets. The resulting data is insufficient to provide a comprehensive evaluation of map balance or design beyond the immediate spawn area.

Objectives Covered

- **Map launch and initial state capture:** Run status: completed.
- **Agent decision logging:** 10 decision event(s) recorded.
- **Movement telemetry:** 22 position sample(s) across 1 cluster(s).
- **Gameplay recording:** /media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/105b84e5-b576-40c1-8e68-4b4e9e472044.mp4

Objectives Omitted

- **Navigate the entire map:** Navigation softlock at starting point.
- **Engage all enemy types encountered:** Blocked by inability to reach starting weapon.
- **Find secrets where accessible:** Agent failed to leave the start area.
- **Stress-test geometry:** Agent stuck on collision geometry at start.
- **Document resource balance issues:** Incomplete testing due to total progression block.

Uncovered attributes: Multiplayer interactions, map-specific scripting, and mid-to-late map geometry were not tested. A surprising observation was the inability of the agent to pathfind to a visually unobstructed shotgun, indicating a potential navigation-mesh or collision-hull issue.

Defects And Risks

A total of 4 defects were found: 1 Critical (softlock), 1 Major (geometry access), and 2 Minor (difficulty flags and unreachable secrets). The most impactful defect is the critical navigation softlock at the player start.

Patterns: All defects appear concentrated at the initial player spawn, where inaccessible geometry prevents the acquisition of essential starting equipment.

Severity	Priority	Type	Tick	Description
1	1	softlock_navigation	5	Run stalled after repeated stuck decisions The run produced repeated stuck events with no meaningful progress. This usually indicates blocked navigation, unreachable objectives, or an agent loop that should be reviewed with the recording and action trace.
2	2	agent_observed_geometry	8	Agent-observed geometry issue [GEOMETRY] At tick 8, position (1001.4, -3228.3): shotgun at object ID 0 is visually reachable but inaccessible. Player impact: unable to acquire starting weapon, combat progression blocked. Severity: critical.
3	2	difficulty_spawn_mismatch	0	Difficulty-specific thing flags hide map content At difficulty 1, 44 of 69 enemies and 7 of 78 items from raw static analysis do not spawn in single-player because of Doom skill or multiplayer flags.
3	3	unreachable_secret	9	Secrets not reached Static analysis found secret sectors, but the agent never entered one.

Risk Areas

- **Player Start Position (1001, -3228):** high Critical collision or geometry issue prevents any player progression.

Activity Log Summary

1) Static analysis phase (Tick 0), 2) Map exploration phase (Ticks 0-9), 3) Combat phase (None), 4) Defect detection phase (Ongoing), 5) Report generation phase.

Exploration was entirely absent due to the navigation block, forcing the agent into an continuous, unsuccessful attempt to reach the starting weapon.

Tick	Event	Position	HP	MCP Action	Stop reason
5	stuck	1001.4, -3228.3	100	move_to	enemy_nearby
6	stuck	1001.4, -3228.3	100	move_to	enemy_nearby
7	stuck	1001.4, -3228.3	100	move_to	enemy_nearby
8	stuck	1001.4, -3228.3	100	move_to	enemy_nearby
9	stuck	1001.4, -3228.3	100	move_to	enemy_nearby

Limitations And Recommendations

The agent was unable to test any doors, lifts, or switches as it was unable to move from the starting position.

Good Quality Areas

- **N/A:** No areas of the map were successfully navigated to warrant a quality assessment.